

ΑΡΙΣΤΟΤΕΛΕΙΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΟΝΙΚΗΣ
ΣΧΟΛΗ ΘΕΤΙΚΩΝ ΕΠΙΣΤΗΜΩΝ
ΤΜΗΜΑ ΜΑΘΗΜΑΤΙΚΩΝ



Semi-Quantum Cryptography

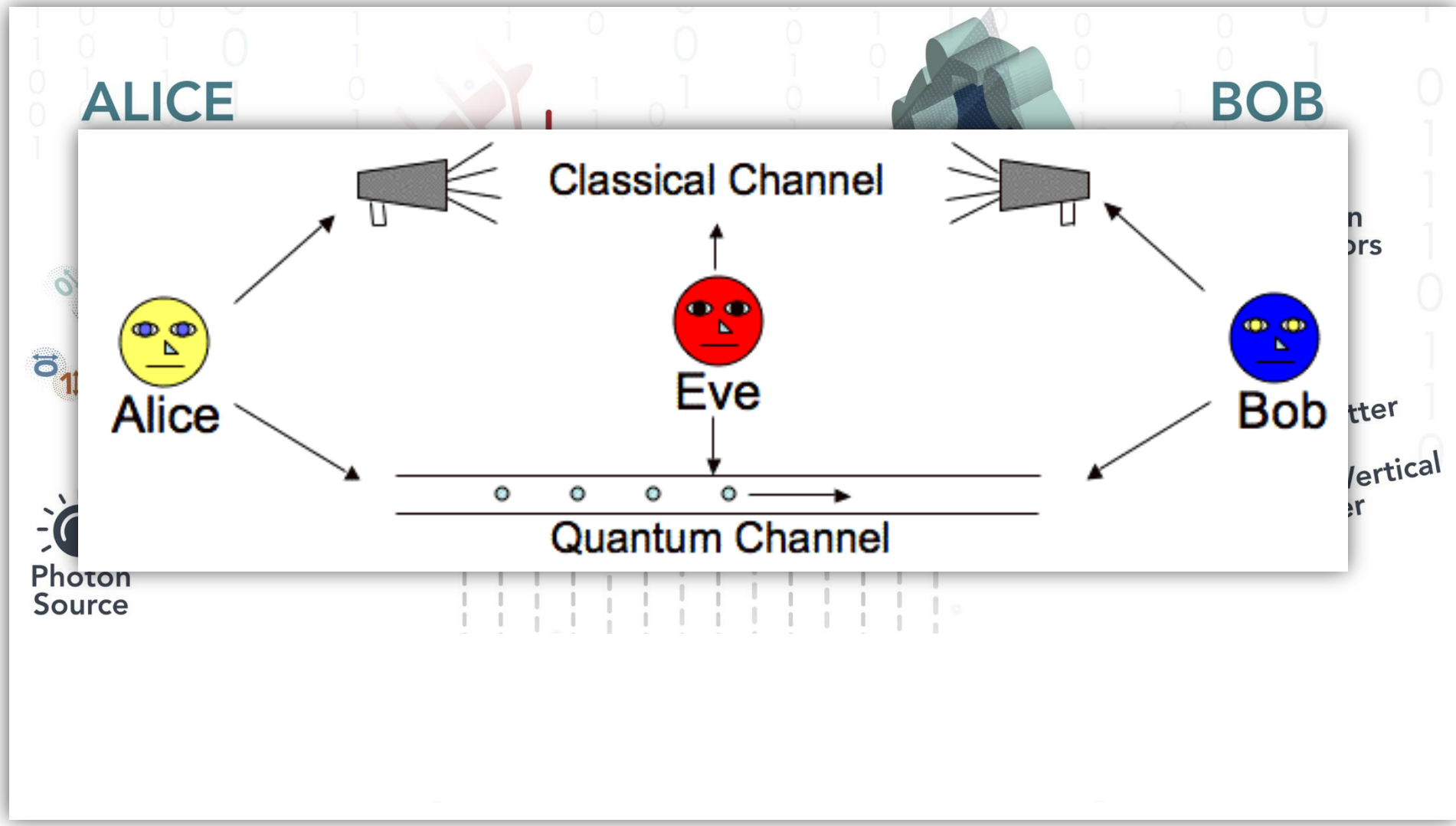
ΦΛΩΡΙΑΣ ΠΑΠΑΔΟΠΟΥΛΟΣ



PRELIMINARIES



Quantum Key Distribution (QKD)



Security Terms

Noise Tolerance	Quantum Channel Quality
	Security Assumptions
	Post-processing Techniques
Robustness	Completely robust
	Partially robust
	Completely nonrobust

A new direction?

- Quantum capable
  User that can manipulate qubits in certain ways

→ Is this necessary for all users?

How quantum must a protocol be to gain advantage over classical communication?

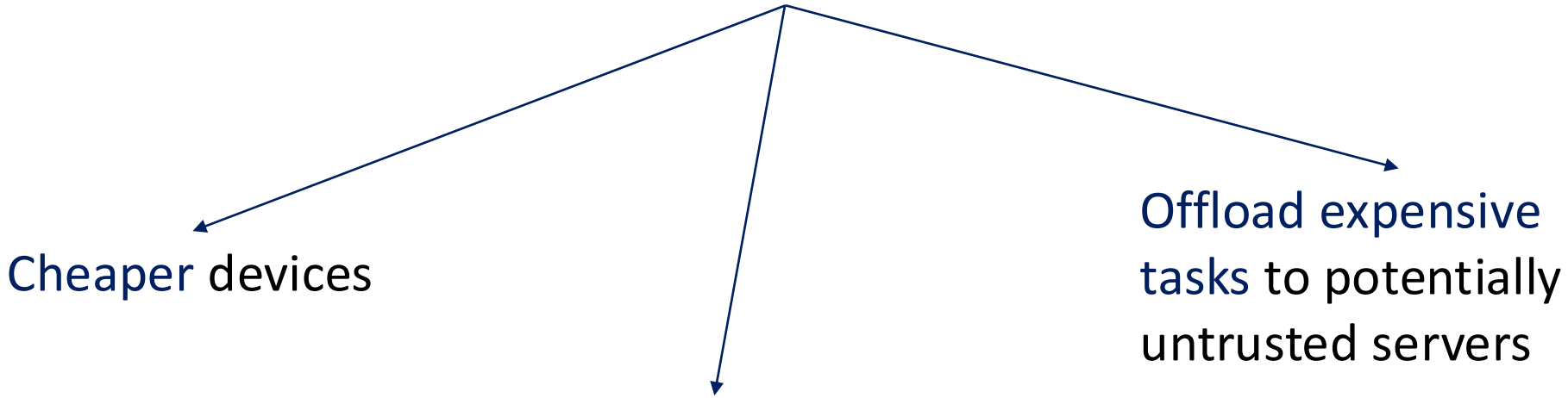
Semi-Quantum communication



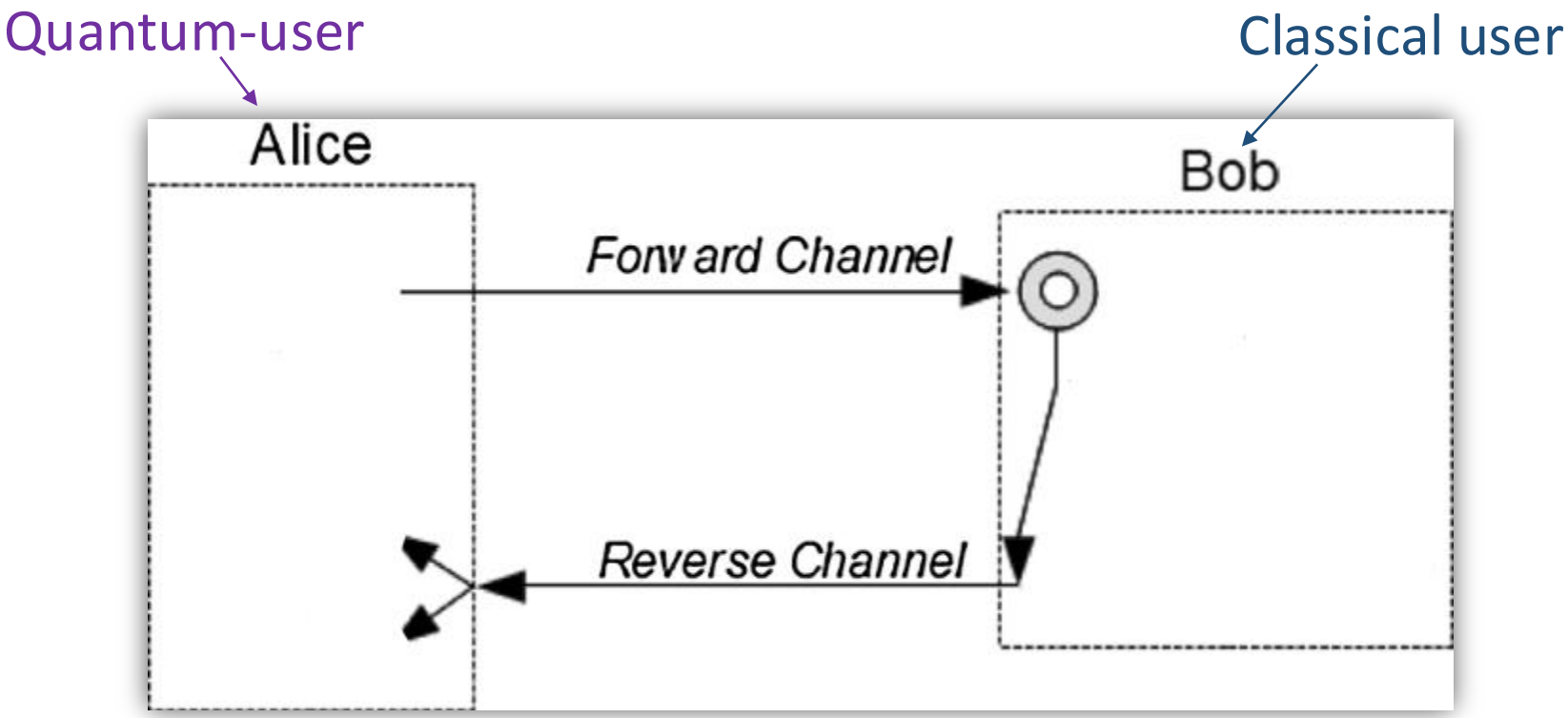
Semi-Quantum Key Distribution



Advantages of Minimizing Quantum Requirements

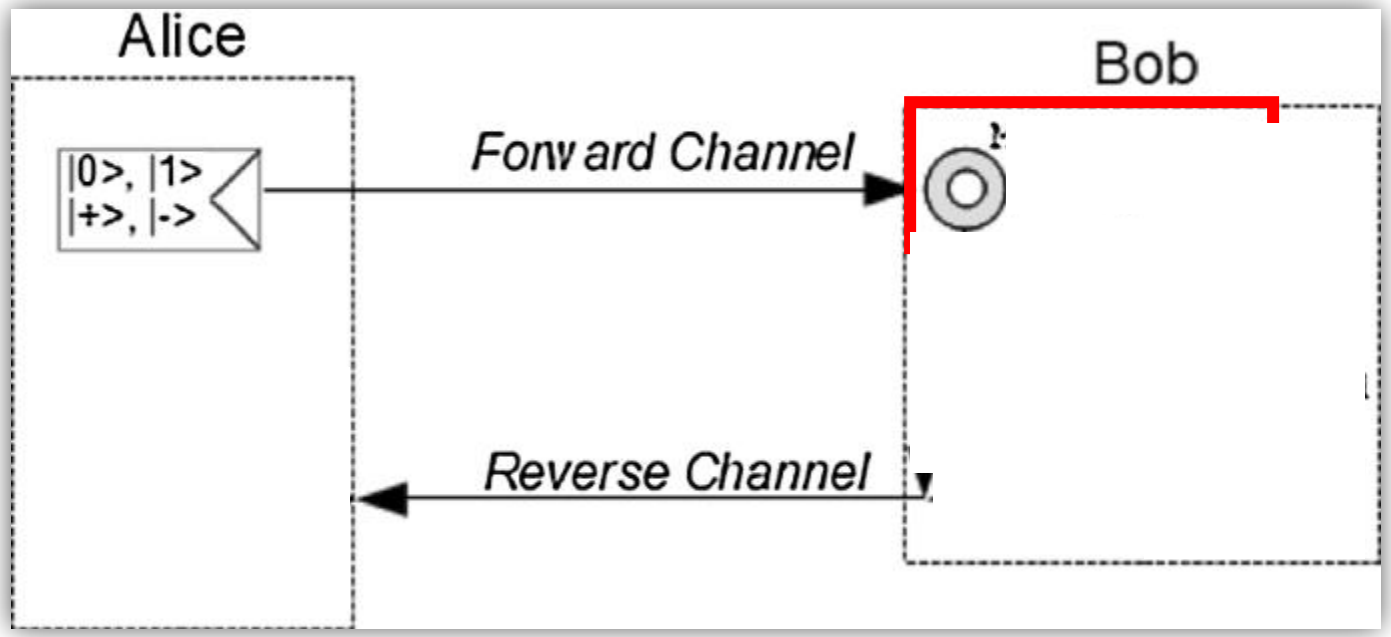


The setup – Channels

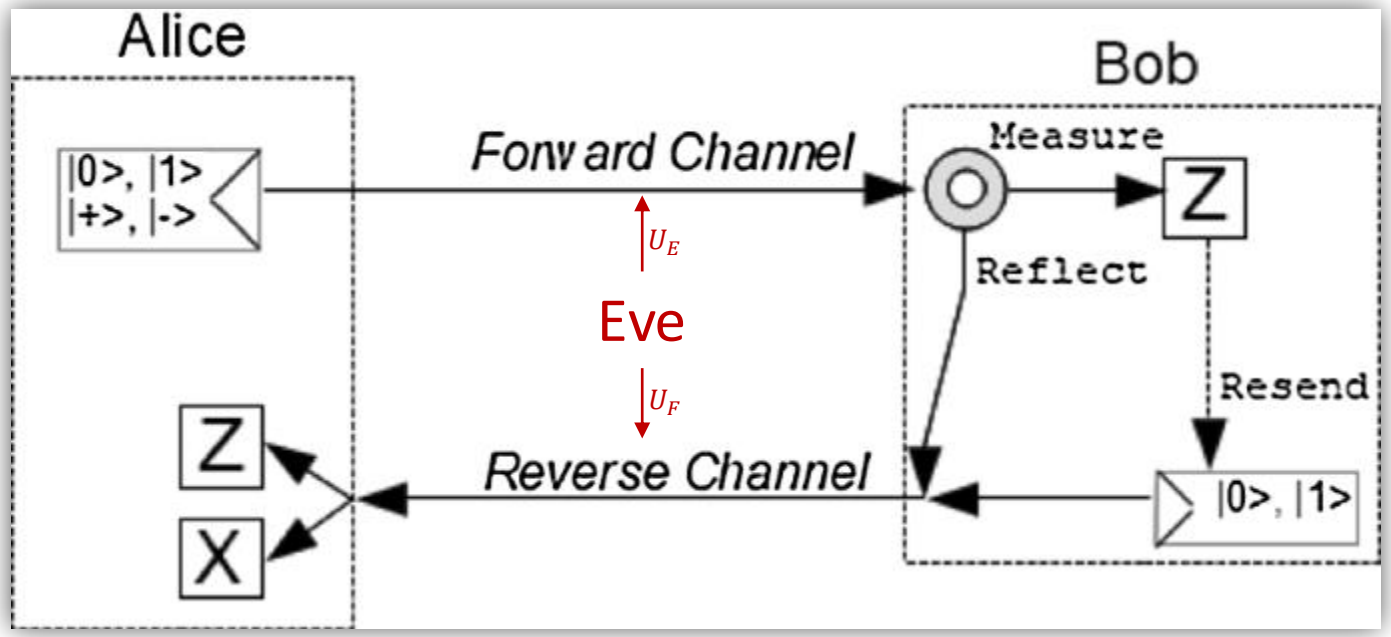


The setup – Classical options

1. Measure
2. Measure & Resend
3. Reflect
4. Permute
5. Prepare



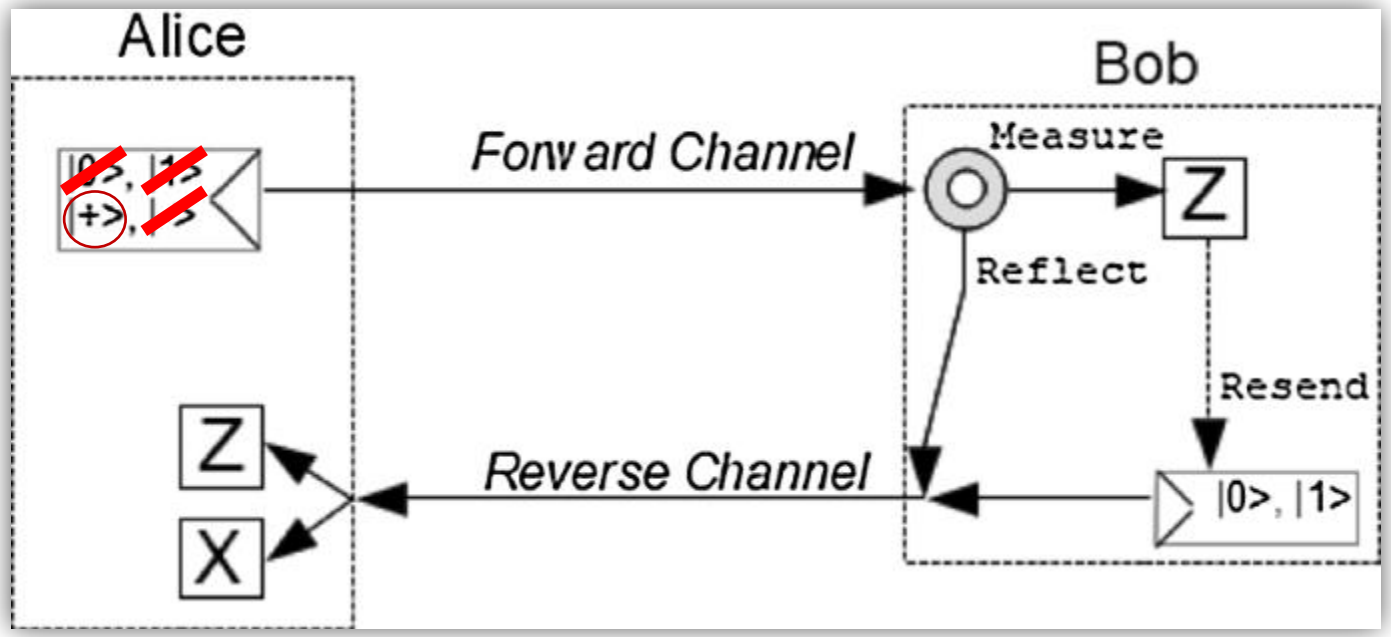
“QKD Protocol With Classical Bob” [3]



- If *Measure* instead of *Measure and Resend* ---> CNOT

New Directions in SQKD – Reducing Requirements

Single State SQKD [4]



- ✓ Completely robust
- ✓ Security properties comparable to fully-quantum

New Directions in SQKD – Others

∃ Protocol where B doesn't *Measure* nor *Prepare* [5]

∃ Protocol without authenticated channel [6]

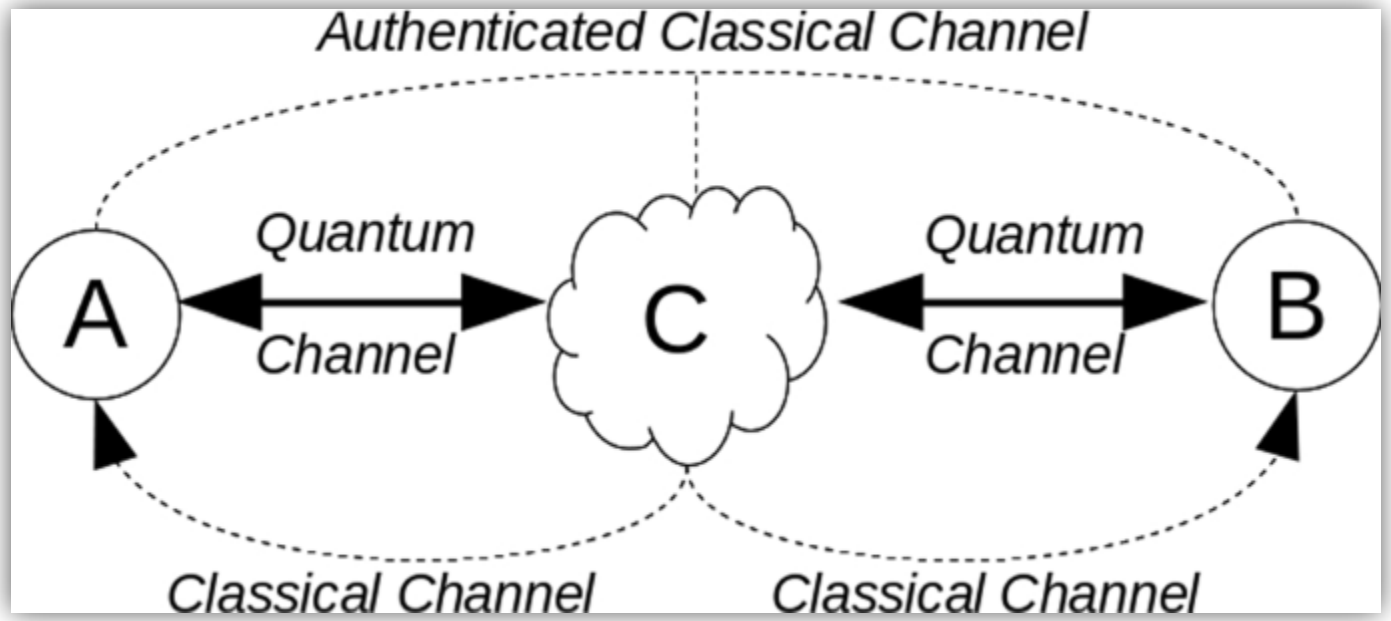
∃ Protocol where A uses data from both channels [7]

∃ Protocols that use high-dimensional states [8]

New Directions in SQKD – Multiuser

Trusted Quantum User [9]

Mediated [10]





Beyond Key Distribution



Applications of Semi-Quantum Communication

Semi-Quantum Secret Sharing (SQSS) with a quantum dealer [11]

Semi-Quantum Secure Direct Communication (SQSDC) [12]

Semi-Quantum Quantum Dialog [13]

Semi-Quantum protocols for Identity Verification [14]

Semi-Quantum Random Number Generation [15]

Semi-Quantum Copy Protection [16]



Practical Semi-Quantum



Major Challenges in Implementation

- Require two-way quantum channel
- *Measure and Resend* problem in practice
- Switching between *Measure and Resend* and *Reflect*
- Device imperfections

Mirror Protocol [16]
Uses controllable mirror
Single state protocol
Completely robust



Ευχαριστώ



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